

Structural Design of a Foil Using Geometric Programming : Modeling and Solution

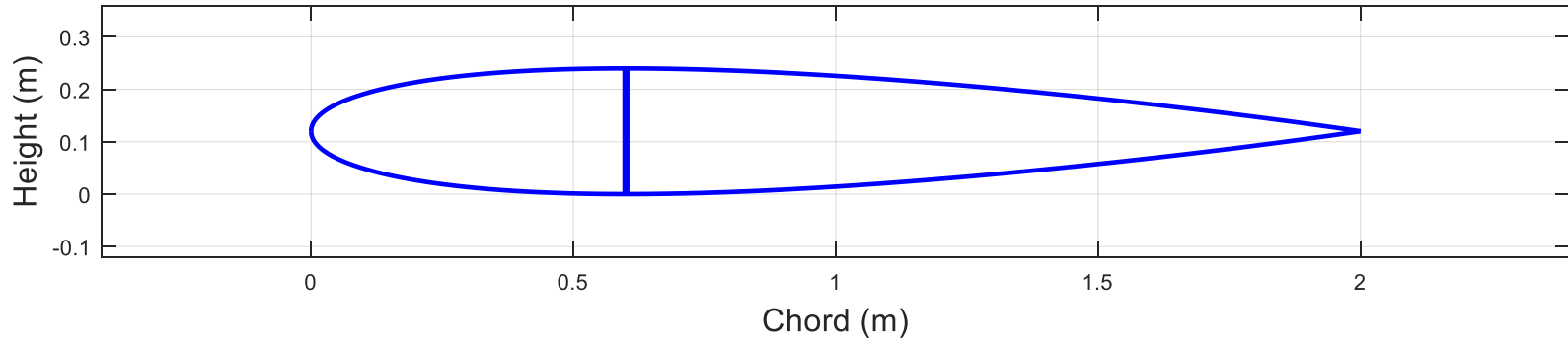
ENEL 641 Optimization for Engineers

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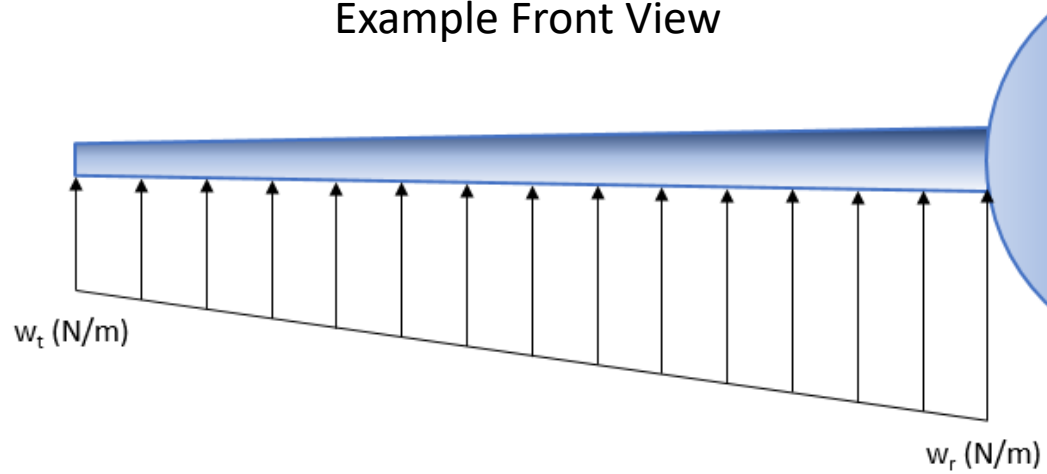
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MODELING

Segment Cross section



Example Front View



MODELING

Basic Form

GP in Convex Form

Minimize: volume

$$\sum A_{xi}(b_i, h_i, t_i) l$$

$$\ln \left[\sum \exp[x_{1i} + x_{2i} + \ln(K_{vol i})] \right] \quad i = 1, \dots, n \quad (1)$$

s.t.: max deflection of tip

$$y_1(b_i, h_i, t_i) \leq y_{max}$$

$$\ln \left[\sum \exp[-3x_{1i} - x_{2i} + \ln(K_{y1 i})] \right] \leq 0 \quad i = 1, \dots, n \quad (2)$$

max stress

$$\sigma_{max i}(b_i, h_i, t_i) \leq \sigma_{max}$$

$$-2x_{1i} - x_{2i} + \ln(K_{sm i}) \leq 0 \quad i = 1, \dots, n \quad (3)$$

max planform area

$$\sum A_i(b_i) \leq A_{max}$$

$$\ln \left[\sum \exp[x_{1i} + \ln(K_{a i})] \right] \leq 0 \quad i = 1, \dots, n \quad (4)$$

monotonic increasing chord

$$b_i \leq b_{i+1}$$

$$x_{1i} - x_{1i+1} + \ln(K_{r i}) \leq 0 \quad i = 1, \dots, n-1 \quad (5)$$

bound on height

$$h_{min} \leq h_i \leq h_{max}$$

$$\ln(h_{min}) \leq x_{1i} \leq \ln(h_{max}) \quad i = 1, \dots, n \quad (6)$$

bound on skin thickness

$$t_{min} \leq t_i \leq t_{max}$$

$$\ln(t_{min}) \leq x_{2i} \leq \ln(t_{max}) \quad i = 1, \dots, n \quad (7)$$

ratio of height to chord

$$\frac{h_i}{b_i} = r_{des i}$$

$$i = 1, \dots, n \quad (8)$$

desired chord at tip

$$b_1 = b_{tip}$$

$$x_{11} - \ln(b_{tip} r_1) = 0 \quad (9)$$

desired skin thickness mid-span

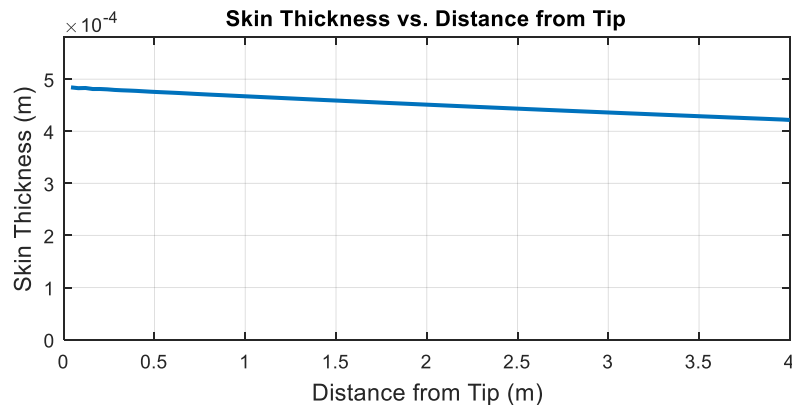
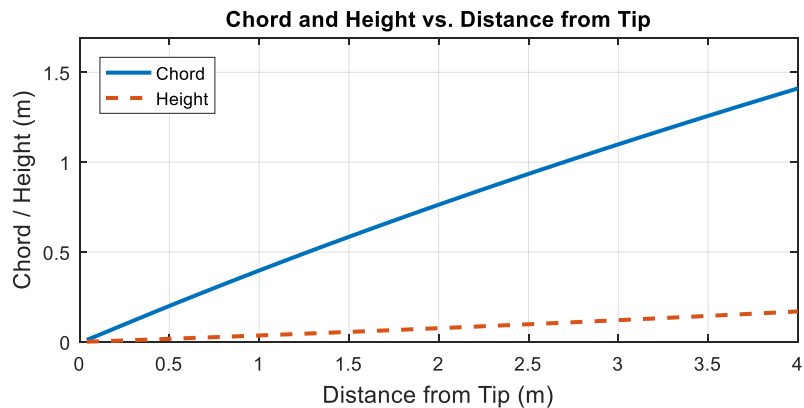
$$t_i = t_{des}$$

$$x_{2i} - \ln(t_{des}) = 0 \quad i = 0.4n, \dots, 0.6n \quad (10)$$

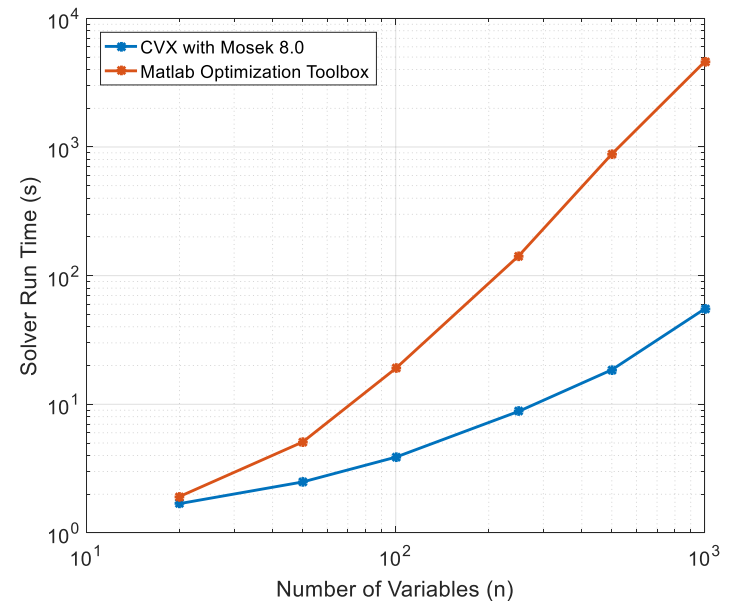
PROBLEM SOLUTION

Case 1

- max deflection 0.1 m



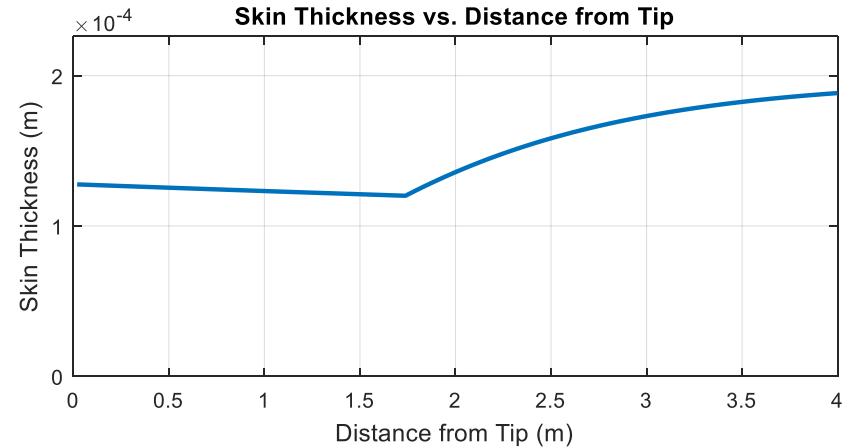
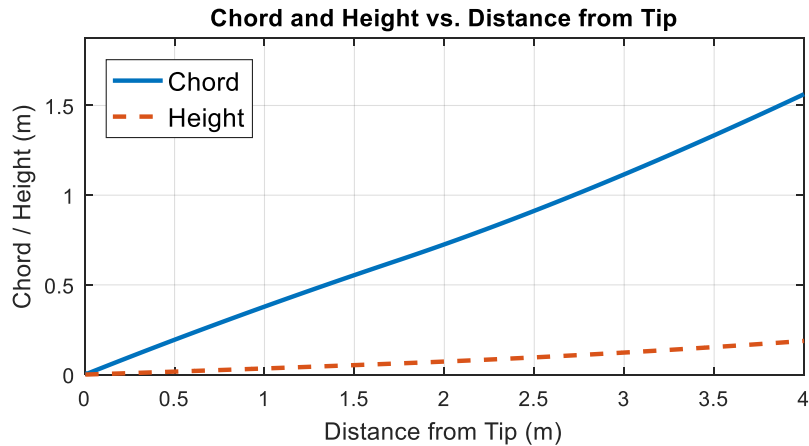
Parameter	Value	Units
General parameters		
Foil span (one side) (L)	4	m
Modulus of Elasticity (E; Al 6061-T6)	68.9	MPa
Load distribution tip (w_t)	0	N/m
Load distribution root (w_r)	1500	N/m
Ratio of spar width to skin thickness (a)	5	d-less
Constraints parameters		
Max deflection (y_1)	0.1	m
Max stress (σ ; Al 6061-T6) *	127	MPa
Max planform area (A)	3	m ²
Min foil height (h_{min})	0.0005	m
Max foil height (h_{max})	0.5	m
Min skin thickness (t_{min})	0.0001	m
Max skin thickness (t_{max})	0.01	m



PROBLEM SOLUTION

Case 2

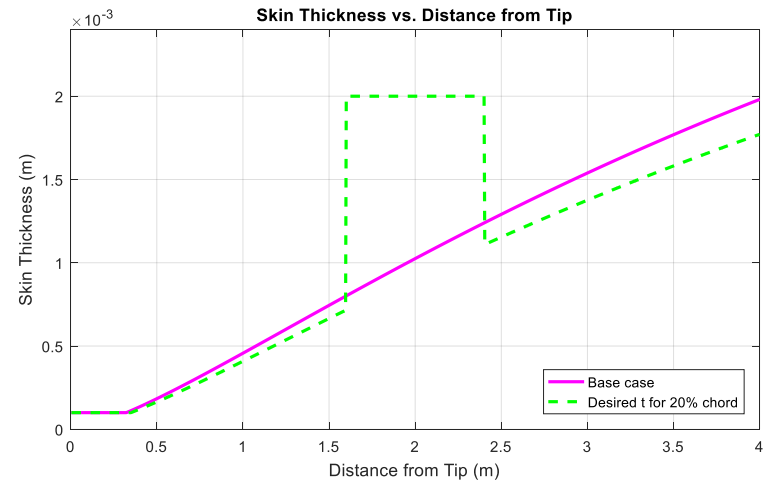
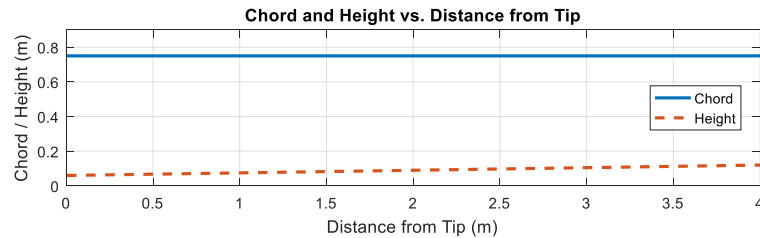
- max deflection 0.3 m
- $n = 1000$ segments
- CVX with Mosek 8.0



PROBLEM SOLUTION

Cases 3, 4

- max deflection 0.1 m
- constant chord of 0.75 m
- $n = 1000$ segments
- CVX with Mosek 8.0



CONCLUSIONS

Foil structural design problem can indeed be modeled as a geometric program

CVX with Mosek 8.0 worked very well for solving several scenarios of the optimization problem with 1000 – 2000 variables